Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	RM15 Refunders
Selected Domain	Domain 2-ALPHA RED
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth (Targets 8.2 and 8.3); related SDG 9
Project / Solution Name	StockLink
Name (Leader)	Wang Yanfeng
Email & Phone (Leader)	Redacted in this public draft. Use the team's private copy for the official submission contact details.
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small-business owners and small teams in Malaysia often track orders and stock separately across WhatsApp, spreadsheets, paper notes and manual follow-ups. When an order is not reflected consistently in inventory, staff may miss low-stock risks, spend time reconciling records, or make avoidable replenishment decisions. These are the team's problem observations and design assumptions; we have not yet conducted interviews or measured a representative baseline. The burden matters because small teams have limited time and operational capacity. StockLink focuses on linking confirmed orders to inventory estimates and showing understandable stock-risk alerts, while leaving decisions with the owner. We will validate the workflow and quantify time and accuracy during Phase 2 user testing.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

StockLink is a lightweight order-and-inventory workflow for Malaysian micro and small businesses. The current public prototype is a demo: users can review sample orders, inventory, deliveries and activity; a deterministic rules-based forecast estimates near-term stock risk from entered stock, confirmed orders, baseline demand, recipe assumptions and expected receipts. It explains why an item is flagged and keeps replenishment decisions under human control. No AI model or external API is currently embedded in the app. For Phase 2, we propose evaluating an Orbyte-approved GenAI feature that could turn an owner-approved message into a draft order for review; it will not create orders or contact suppliers without confirmation. The team has completed the mandatory GenAI, database and cloud workshops. The current build is a functional, deployed prototype, not evidence of measured business impact.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

UI (React 19.1.0, TypeScript) → static build (Vite 7.3.7) → browser-local demo state (localStorage) → deterministic rules-based stock estimate and explainable alert → owner reviews and confirms changes → local activity log. No backend, database, authentication, external API, or live POS/supplier connection is present. Public static deployment: Replit Static Publish. Sponsor resource: Replit voucher supplied by the hackathon; mentor guidance. Future integrations/AI are proposed only and require validation and user approval.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We used the team's stated workflow problem to scope a simple prototype around order confirmation, inventory risk and owner review. The deployed demo exposes Overview, Orders, Inventory, Deliveries and Activity pages, with sample records and explainable alerts. It is technically viable as a static front end and keeps current decisions visible to the user. We have not interviewed users or completed usability testing, so ease of learning, forecast accuracy and time savings remain unvalidated. In Phase 2 we will test representative tasks with Malaysian micro/small-business users, record baseline and completion time, error rates and alert usefulness, then revise the interface and document integration, privacy and deployment needs before connecting real systems.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2: improve productivity through technological upgrading and innovation. SDG 8.3: support entrepreneurship and the growth of micro, small and medium enterprises. Related: SDG 9 (innovation and infrastructure). Source: UN DESA, https://sdgs.un.org/goals/goal8
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Not measured yet. The team has no interview, timing, error-rate or stock-out dataset, so no operational baseline is claimed. Phase 2 baseline: record median minutes and order/stock mismatches across a defined pilot before intervention.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: median minutes to reconcile a confirmed order with inventory; also track order-to-stock mismatch rate and stock-risk alert precision during a pilot.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	[PROPOSED] Aim for at least 30% lower median reconciliation time in a small, defined pilot, without increasing mismatch rate. This is a test target, not an achieved result; finalize after baseline measurement.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

✓	All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
✓	The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
✓	The proposed solution targets a functional, deployed application rather than a theoretical business case study.
✓	All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓No

No pre-existing code/design/research reported by the team.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✓Yes ☐ No

Third-party open-source libraries are used; principal runtime/build libraries are listed below. See their license notices and repository links.

Name	Type (code/dataset/model)	License
React 19.1.0 / React DOM 19.1.0	Code/framework	MIT — https://github.com/facebook/react
Vite 7.3.7	Build tool	MIT — https://github.com/vitejs/vite
Tailwind CSS 4.3.3	CSS framework	MIT — https://github.com/tailwindlabs/tailwindcss
Recharts ^2.15.2	Charts	MIT — https://github.com/recharts/recharts
Lucide React 0.545.0	Icons	ISC — https://github.com/lucide-icons/lucide
Wouter ^3.3.5	Routing library	Unlicense — https://github.com/molefrog/wouter

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓No

No employment, internship or consulting IP restrictions reported by the team.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature： 

Date：24.10.2026